CO543 / CO5430
Computer Vision Project - 2026

Vehicle or Pedestrian Tracking in Video

M1: Proposal and Project Plan

Group Number	G07
Project ID	P07
Project Title	Vehicle or Pedestrian Tracking in Video
Project Category	Object Tracking
Dataset Source	MOT16 - https://www.kaggle.com/datasets/takshmandar/mot16-dataset
GitHub Repository	https://github.com/cepdnaclk/e23-co5430-vehicle-pedestrian-tracking.git

Group Members

#	Name	Registration No.
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1. Problem Statement and Motivation

Multi-object tracking (MOT) in video — following vehicles and pedestrians across frames while preserving their identities — is a core perception task for intelligent transportation systems, traffic monitoring, and autonomous driving. The problem is challenging because targets frequently occlude one another, enter and leave the scene, change scale and appearance under camera motion, and move through cluttered urban backgrounds.

The objective of this project is to build a tracking-by-detection pipeline that (a) detects vehicles and pedestrians in each video frame, and (b) associates these detections across frames into consistent identity trajectories, while minimising identity switches and track fragmentation. We will start from a simple motion-based baseline tracker and progressively improve it with an appearance-aware association method, evaluating both quantitatively (MOTA, IDF1, ID switches) and qualitatively (visualised tracks).

2. Related Work

- Bewley et al., "Simple Online and Realtime Tracking" (SORT, 2016) — a lightweight tracking-by-detection baseline combining Kalman-filter motion prediction with the Hungarian algorithm for IoU-based data association. We use this as our baseline tracker.
- Wojke et al., "Simple Online and Realtime Tracking with a Deep Association Metric" (DeepSORT, 2017) — extends SORT with a learned appearance embedding (Re-ID network) to reduce identity switches under occlusion. This informs our improved/adapted method.
- Zhang et al., "ByteTrack: Multi-Object Tracking by Associating Every Detection Box" (2022) — improves association robustness by also matching low-confidence detection boxes instead of discarding them, which is particularly useful in the crowded scenes present in MOT16.
- Milan et al., "MOT16: A Benchmark for Multi-Object Tracking" (2016) — defines the dataset, annotation format, and the standard MOTA/MOTP/IDF1 evaluation protocol that this project will adopt.

Our approach differs from a direct re-run of these methods by combining a modern pretrained detector (e.g., YOLOv8) with a comparative study of the classical SORT baseline versus an appearance-augmented association scheme, and by analysing performance specifically on the vehicle vs. pedestrian sub-classes within MOT16.

3. Dataset

Dataset source: MOT16 (Multiple Object Tracking Benchmark 2016), obtained via Kaggle: https://www.kaggle.com/datasets/takshmandar/mot16-dataset. MOT16 contains 14 video sequences (7 train / 7 test) filmed from both static and moving cameras in unconstrained, crowded urban environments, with manually annotated bounding boxes and identity labels for pedestrians and vehicles.

Preprocessing plan: convert MOT16's frame-and-annotation format into a per-sequence detection/track structure; use the official 7 training sequences (which have ground truth) for our own train/validation split, holding out 1–2 full sequences as a held-out validation set (since MOT16 test sequences have no public ground truth, we will report final metrics on our held-out split). Frames will be resampled/resized as needed to fit detector input constraints, and class labels will be filtered/mapped to two categories of interest: pedestrian and vehicle.

Licensing/ethics: MOT16 is a publicly released academic benchmark distributed for research use; it contains pedestrians in public street footage but no personally identifying metadata beyond bounding boxes. We will cite the original benchmark paper and Kaggle mirror, and will not redistribute the raw dataset in our GitHub repository — only download instructions and small sample frames for demo purposes.

4. Proposed Method

Baseline: A pretrained object detector (YOLOv8, used without fine-tuning) generates per-frame bounding boxes for vehicles and pedestrians. These are fed into SORT, which uses a Kalman filter to predict object motion and the Hungarian algorithm with IoU cost to associate detections to existing tracks across frames. This gives a naive, motion-only tracking-by-detection baseline.

Improved/adapted method: We will augment the baseline with an appearance-based association step (a DeepSORT-style Re-ID embedding and/or ByteTrack-style low-confidence box recovery) to better handle occlusions and reduce identity switches, and compare classical IoU-only association against the appearance-augmented version as our required ablation/comparison.

Minimum goal: a working detector + SORT baseline producing tracked bounding boxes and trajectory visualisations on at least one MOT16 sequence.

Expected goal: baseline plus an appearance-augmented tracker, evaluated quantitatively on multiple sequences with a clear comparison table.

Stretch goal: real-time inference demo (webcam or recorded traffic video) with a lightweight tracker configuration, and separate performance breakdown for vehicles vs. pedestrians.

5. Evaluation Plan

We will use the standard MOT evaluation protocol via the py-motmetrics toolkit, reporting: MOTA (Multiple Object Tracking Accuracy), MOTP, IDF1, number of ID switches, and track fragmentation, computed on our held-out validation sequence(s). Qualitative evaluation will include annotated video frames/GIFs showing tracked trajectories, with a specific discussion of failure cases (e.g., occlusion-induced ID switches, missed detections at small scale, and false positives in cluttered backgrounds). The required baseline-vs-improved-method comparison (SORT vs. appearance-augmented tracker) will be the central ablation of the project.

6. Timeline

Milestone	Date	Deliverable / Focus	Owner(s)
M0 – Group & topic registration	1–7 Jul 2026	Group formed (G07), topic P07 selected, dataset & GitHub link registered	All members
M1 – Proposal & project plan	14 Jul 2026	This proposal document	All members
Data preparation	15–21 Jul 2026	Download MOT16 sequences, verify annotations, build data-loading & split scripts	Rahman, Nular
M2 – Dataset & baseline checkpoint	28 Jul 2026	Baseline detector + SORT tracker running end-to-end on a subset	Piraveen, Thanush
Method development	29 Jul–15 Aug 2026	Integrate appearance re-ID (DeepSORT) / ByteTrack-style association; tune parameters	All members
M3 – Prototype & preliminary results	18 Aug 2026	Preliminary MOTA/IDF1 results, qualitative tracks, failure-case analysis	All members
M4 – Experiment freeze	25 Aug 2026	Final experiment plan locked; ablation studies completed	Rahman, Piraveen
M5 – Draft report	1 Sep 2026	Report skeleton, result tables/figures, demo checklist, repo cleanup	Nular, Thanush
M6 – Final submission & demo	7 Sep 2026	Final report, repository, slides, demo video	All members

7. Risks and Mitigation

Risk	**Likelihood / Impact**	**Mitigation**
ID switches increase in crowded scenes with heavy occlusion	High / Medium	Use appearance embeddings (Re-ID) in addition to motion (Kalman filter) for association; tune IoU/cosine-distance thresholds
Limited GPU/compute resources for training or fine-tuning detectors	Medium / Medium	Use pretrained detectors (e.g., YOLOv8/Faster R-CNN) without full retraining; use Google Colab for extra GPU time
MOT16 ground-truth format / evaluation tooling (py-motmetrics) has a learning curve	Medium / Low	Start integration early during the data-preparation phase; follow official MOTChallenge devkit conventions
Uneven workload distribution across 4 members	Low / Medium	Weekly sync meetings, task board on GitHub Projects, clear ownership per milestone (see timeline)
Scope creep beyond feasible timeline	Low / Medium	Fixed minimum/expected/stretch goals defined below; M4 experiment freeze enforced strictly